

Table 1: Representative comparison for vector-final-state radiative widths. Widths are in keV. The last column gives the present calibrated LCSR result with the preferred lattice photon normalization and fixed $\theta = 35.3^\circ$; percentile intervals are shown in square brackets.

Channel	[1]	[2]	[3]	[4]	[5]	[6]	[7–9]	[10]	This work
$D_{s1}(2536) \rightarrow D_s^* \gamma$	5.6	14.6–22.8	9.21	8.9	2.96–3.02	10.4 ± 1.0	...	29	23.0 [16.5, 35.1]
$D_{s1}(2460) \rightarrow D_s^* \gamma$	5.5	14.0–25.1	17.4	15.5	4.74–4.79	...	...	104	92.3 [59.7, 138.7]
$B_{s1}(5750) \rightarrow B_s^* \gamma$	...	...	...	...	...	...	56, 36.9, 39.5	20	10.2 [6.63, 16.24]
$B_{s1}(5830) \rightarrow B_s^* \gamma$	...	...	...	...	...	...	53, 57.3, 98.8	41	7.46 [3.30, 24.0]

For the bottom rows, the combined literature column lists, in order, [7], [8], and [9]. The machine-readable CSV stores the θ -scan and legacy-photon alternatives.

References

- [1] Stephen Godfrey. Properties of the charmed P-wave mesons. *Phys. Rev. D*, 72:054029, 2005.
- [2] Jose L. Goity and Winston Roberts. Radiative transitions in heavy mesons in a relativistic quark model. *Phys. Rev. D*, 64:094007, 2001.
- [3] N. Green, W. W. Repko, and S. F. Radford. Note on predictions for $c\bar{s}$ quarkonia using a three-loop static potential. *Nucl. Phys. A*, 958:71–86, 2017.
- [4] Stanley F. Radford, Wayne W. Repko, and Michael J. Saelim. Potential model calculations and predictions for $c\bar{s}$ quarkonia. *Phys. Rev. D*, 80:034012, 2009.
- [5] S. F. Chen, J. Liu, H. Q. Zhou, and D. Y. Chen. Electric transitions of the charmed-strange mesons in a relativistic quark model. *Eur. Phys. J. C*, 80(4):290, 2020.
- [6] J. G. Korner, D. Pirjol, and K. Schilcher. Radiative decays of the p wave charmed mesons. *Phys. Rev. D*, 47:3955–3961, 1993.
- [7] Qi Li, Rong-Hua Ni, and Xian-Hui Zhong. Towards establishing an abundant B and B_s spectrum up to the second orbital excitations. *Phys. Rev. D*, 103(11):116010, 2021.
- [8] Stephen Godfrey, Kenneth Moats, and Eric S. Swanson. B and B_s meson spectroscopy. *Phys. Rev. D*, 94(5):054025, 2016.
- [9] Qi-Fang Lu, Ting-Ting Pan, Yan-Yan Wang, En Wang, and De-Min Li. Excited bottom and bottom-strange mesons in the quark model. *Phys. Rev. D*, 94(7):074012, 2016.
- [10] A. E. Bondar and A. I. Milstein. Phenomenology of D_{s1} mesons radiative transitions. *Phys. Rev. D*, 111(11):114019, 2025.